Hynes Birmingham

Storrs, CT • (860) 839-5859 • hynes.birmingham_ii@uconn.edu • linkedin.com/in/hmbirmingham • github.com/hmbirmingham • hmbirmingham.me

EDUCATION

University of Connecticut — Storrs, CT
B.S. Biomedical Engineering | Minor in Computer Science, Math, Electronic Systems | Concentration: Systems & Instrumentation — *May 2029*
CAPS Research Scholar (14th Cohort) — **GPA**: 3.41

TECHNICAL EXPERIENCE

Ryberg Lab — University of Connecticut, Allied Health Sciences — Storrs, CT (Hybrid)
Undergraduate Researcher — CAPS Research Scholar — *Nov 2025 – Present*

- Designed and built an automated bacterial colony counting instrument: embedded Linux deployment on Raspberry Pi 5, custom KiCad power-control PCB (designed and ordered), and a Python/OpenCV computer vision pipeline achieving 96.3% count accuracy and F1 0.957 on synthetic plate validation set (820 plates, 13,690 ground-truth colonies)
- Engineered a continuous self-validating testing pipeline: 59/59 unit tests passing, synthetic data generation, automated execution, and provenance-tracked reporting (git commit hash, environment snapshot, input hash) across 6,345 SLOC
- Built a two-layer anomaly detection system combining a statistical detector (Z-score/IQR) with a Random Forest classifier; pipeline processes plates in 62.3 ± 0.8 ms with a false-positive rate below 0.10% on non-dense conditions
- Structured 39-organism × 11-media-type profile system with per-run provenance tracking; accepted to present original research at UMBC Undergraduate Research Conference (Sep 2026)

TECHNICAL PROJECTS

STM32 Bare-Metal GPIO Driver — Personal Project — Jul 2026 – Present

- Wrote a register-level GPIO driver for the STM32F411RE in C with no HAL abstraction, directly configuring 9 MMIO registers across 3 peripherals (RCC, GPIOA, USART2) via memory-mapped I/O; flash image 752 B at -O0; signal timing verified with logic analyzer

FocusGuard — Multi-Modal Sensor Fusion System — 2026 – Present

- Architected a multi-threaded producer-consumer system in Python fusing real-time computer vision (MediaPipe face mesh) with keyboard behavioral signals; 26 ms median sensor-to-score latency at 30 fps, 116 MB steady-state RSS, zero crashes over 6-minute soak test (10/10 unit tests passing, 1,008 SLOC)
- Implemented thread-safe lock-guarded shared state across independently-clocked sensor threads; Flask REST API and live dashboard serving real-time and historical data

Automated Bacterial Colony Counter — Full Stack Instrument — Feb 2026 – Present

- End-to-end Flask API: 71.1 ± 24.6 ms POST-to-SSE response (image quantification, annotated JPEG, anomaly scoring, CSV logging); 10,600+ req/s sustained throughput on read endpoints; 0 failures across 120 timed API runs

HuskyBook — Peer-to-Peer Campus Directory — Aug 2026 – Sep 2026

- Full-stack web app: Node.js + Express REST API (16 endpoints, 5-table SQLite schema), Vite front end; 0.07–0.16 ms read latency, 10,600 req/s throughput; 32/32 tests passing, 1,724 SLOC

LEADERSHIP

Vergnano Institute of Impact, University of Connecticut — Storrs, CT
BRIDGE STEM Coach — Jun 2026 – Aug 2026

- Coached 43 incoming first-generation and underrepresented STEM students through UConn's BRIDGE transition program
- Independently designed and delivered a biotechnology and instrumentation module to a cohort of 8 students covering bacterial transformation, fluorescence imaging, and quantitative data analysis
- Built a companion Python tool in Google Colab enabling students without local software access to fully process and visualize their own experimental data

Scholars House, University of Connecticut — Storrs, CT
Learning Community Tutor — Fall 2025 – Spring 2026

- Provided STEM tutoring to residents of Scholars House, a living-learning community for Black men in higher education; 2–3 sessions/week over 12 weeks, contributing to the community's 3.0 GPA goal by Spring 2026

Innovation Zone — University of Connecticut — Storrs, CT
Students as Makers (SAMs) — Founding Cohort — *Feb 2025 – Dec 2025*

- Member of the founding cohort of UConn's Students as Makers program; fabricated 7 projects for improvement of the Innovation Zone and related services utilizing 3D printing, manufacturing, and power tools, saving over $2,300 compared to purchasing equivalent equipment or products

Non-Technical Work Experience

Jersey Mike's Subs — West Hartford, CT
Shift Manager — May 2025 – Present

- Process $4,000+ in weekly cash sales, manage inventory and shift staff, and report to the general manager and franchise stakeholders as needed.

SKILLS

Embedded & Hardware: C, STM32/ARM Cortex-M4, register-level MMIO drivers, ARM GCC, st-flash, KiCad PCB design, embedded Linux (Raspberry Pi OS, systemd, cross-compile), logic analyzer
Software & Systems: Python, C, MATLAB, LabView, OpenCV, Flask/REST APIs, multi-threaded systems, scikit-learn, pandas, NumPy, Node.js, Git
Data & Validation: ETL pipelines, SQLite, automated testing, provenance tracking, structured data logging, OOP design